

Gioia Zheng

NLP AND RESEARCH ENGINEERING | INFORMATION RETRIEVAL | RAG EVALUATION

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Profile

Final-year Applied Computer Science and AI student and former Microsoft NLP Algorithm Engineer Intern. Combines end-to-end ML engineering with research-oriented evaluation: builds retrieval-to-generation systems, runs controlled paired studies, and ships traceable artifacts with tests, CI, and failure analysis. Bilingual in Italian and Chinese, with professional English and experience across Italian and Chinese environments.

6,980	0.197 to 0.368	40	2
paired queries evaluated	Token-F1 improvement	real SciFact traces	merged upstream changes

Education

Sapienza University of Rome

B.Sc. in Applied Computer Science and Artificial Intelligence

Sep 2022 - Expected Dec 2026

Experience

NLP Algorithm Engineer Intern, Microsoft (China)

Remote | [Public project repository](#)

Feb 2026 - May 2026

- Developed and evaluated an end-to-end question-answering pipeline on MS MARCO passage data using BM25, FAISS dense retrieval, cross-encoder reranking, and T5 generation.
- Evaluated 6,980 paired queries; Token-F1 increased from 0.197 to 0.368, with a 10,000-sample paired-bootstrap confidence interval of [0.163, 0.178] for the improvement.
- Added schema-versioned experiment manifests, validation checks, controlled comparisons, and failure analysis so results remain traceable to code, data, configuration, and environment.

Web Development Intern, Idearia

Mar 2026 - Jun 2026

- Built two client web tools for Excel-data analysis and employee leave management.
- Maintained PHP/WordPress sites and created minisites from scratch with domain setup, automated email flows, and two n8n workflows.

Selected Projects

RAG Observatory

[GitHub](#) | [Live demo](#)

- Built a trace-based toolkit for inspecting retrieval, context selection, answers, citations, evaluation signals, and failure labels across RAG runs.
- Implemented OpenTelemetry/OpenInference ingestion, stage-aware diagnostic reports, run comparisons, and configuration-sensitive failure analysis.
- Validated a real retrieval-depth comparison on 40 BEIR SciFact traces and maintained 107 automated tests plus type checks, linting, and container CI.

CiboCompass

[GitHub](#)

Archived engineering project

- Built a cross-platform React Native/Expo client with a Go REST API and SQLite persistence for culturally aware food exploration.
- Implemented local caching, a persistent retry queue, and idempotent rating submission; exercised delivery under simulated connectivity drops.

Open Source, Leadership and Awards

Merged Contributions

MTEB ecosystem

- Documented the model-evaluation request workflow in [MTEB #4861](#).
- Added model language-scope metadata, tests, and accessibility coverage in [MTEB Leaderboard #31](#).

Young Representative, UPI Emilia-Romagna

Sep 2022 - Oct 2023

Analyzed school-dropout data and contributed to policy-oriented reports.

1st Place, WISWUP Business Plan Competition

Sep 2021

Co-led a gamified sustainability-app concept and pitched it to a professional jury.

Technical Skills

Programming

Python, Go, SQL

IR and ML

BM25, FAISS, PyTorch, cross-encoder reranking, T5, RAG evaluation

Research Evaluation

Paired bootstrap, confidence intervals, controlled ablations, grounding checks, failure analysis

Software Systems

Linux, Docker, GitHub Actions, REST APIs, SQLite, offline retry queues

Languages

Chinese (Native), Italian (Bilingual), English (Professional), German (Beginner)